Project Report on

SKILL BRIDGE: A Smart Web For Personalized And Secure Education

Submitted in partial fulfilment of the requirements

of the degree of

Bachelor of Engineering in Computer Science and Engineering (AI &ML)

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Declaration

I declare that this written submission represents my ideas in my own words and where other's ideas or words included; I have adequately cited and referenced the original sources. I also declare that I have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in my submission. I understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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Abstract

SKILLBRIDGE The introduction chapter explains the need for a smart education and mentorship system in today's digital world. It highlights the major challenges faced by learners, such as lack of personalized guidance, high mentorship costs, geographical barriers, and language limitations. The SKILLBRIDGE platform is introduced as a solution that uses Artificial Intelligence, real-time communication, and multilingual support to connect learners with industry experts. It also discusses the motivation behind the project, which is to reduce the guidance gap and provide a structured, efficient, and inclusive learning environment. The objectives focus on building an intelligent system that supports real-time mentorship, personalized learning paths, and improved career decision-making.

Furthermore, analyzes existing research and technologies related to AI-based mentorship systems and online learning platforms. It reviews approaches such as MOOCs, adaptive learning systems, and recommendation-based mentor matching. While these systems provide useful features, they often lack integration, personalization, and real-time decision support. The literature survey identifies key limitations like poor mentor matching, lack of multilingual accuracy, and fragmented system design. SKILLBRIDGE addresses these gaps by offering a unified platform that combines AI-driven recommendations, real-time communication, and multilingual translation to deliver a more effective and complete mentorship solution.

SKILLBRIDGE The design methodology chapter explains the architecture and working of the SKILLBRIDGE system. It describes how different modules such as user interface, AI processing layer, database, and external services are integrated. The system collects user data like skills, goals, and queries, and processes it using AI techniques such as vectorization and similarity-based matching to find the best mentor. It also includes features like a real-time AI assistant, multilingual translation using Transformer models, and low-latency audio streaming for live interaction. The chapter highlights the use of technologies like Next.js, Python, PostgreSQL, and AI APIs to build a scalable and

efficient system.

presents the performance and outcomes of the SKILLBRIDGE system after implementation. It shows that the platform successfully provides accurate mentor matching with high efficiency (around 96.2% accuracy). The results also demonstrate effective real-time query resolution through the AI assistant and smooth multilingual communication. Features like dashboards, progress tracking, and learning resources improve user engagement and productivity. The discussion highlights challenges such as data variability and translation accuracy, and explains how they were resolved using advanced techniques like normalization and optimized training. Overall, the system proves to be reliable, scalable, and user-friendly.

summarizes that SKILLBRIDGE is a powerful and intelligent platform that improves traditional learning by combining AI, real-time mentorship, and multilingual support. It successfully reduces the skill gap and provides personalized career guidance. The system is scalable, efficient, and accessible to users from different backgrounds. The future scope includes advanced features like blockchain-based certification, VR/AR-based learning, emotion detection, offline AI support, and job matching integration. These enhancements aim to make the platform more intelligent, inclusive, and closer to real-world industry applications.

Acknowledgements

We would like to express our sincere gratitude to all those who have supported and guided us throughout the successful completion of this project. This work marks an important milestone in our academic journey, and it would not have been possible without the constant encouragement and assistance we received. We are especially thankful to our respected guide, Professor Vaishali Shimpi CSE (AI & ML), VIVA Institute of Technology, Virar, for her valuable guidance, insightful suggestions, and continuous support. Her expertise and mentorship played a significant role in shaping this project.

We would like to convey our heartfelt thanks to Dr. Arun Kumar, Principal of VIVA Institute of Technology, for providing us with this opportunity and necessary support. We are also grateful to all the faculty members of the CSE (AI & ML) department for their guidance and cooperation. Finally, we express our deepest gratitude to our parents, family members, friends, and well-wishers for their constant motivation, encouragement, and moral support, which helped us successfully complete this project.

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Chapter 1: Introduction

1.1 Importance of project

Education and continuous skill development play a vital role in the global economic landscape and support the career progression of a large portion of the professional population [1][2]. However, despite the widespread availability of online learning resources, the educational sector continues to face numerous challenges such as a lack of personalized guidance, high costs of expert mentorship, geographical limitations, and significant language barriers [2][3]. Traditional self-learning practices rely heavily on static content and pre-recorded videos, which often result in a lack of real-time feedback and reduced learning efficiency [3][4]. These limitations highlight the urgent need to integrate modern technological solutions into the mentorship process to improve accessibility, accuracy, and professional growth [4][5].

The **SkillBridge** project addresses these challenges by introducing a smart, technology-driven platform that enhances the mentorship experience through real-time interaction and data-driven matching [5][6]. By leveraging technologies such as Artificial Intelligence (AI), Neural Machine Translation, and low-latency audio streaming, the system enables learners to connect with the most compatible industry experts globally [6][7]. This facilitates the intelligent matching of mentors and mentees based on specific skill gaps and learning styles, allowing for timely intervention and minimizing the risk of career stagnation [7][8]. Additionally, the system promotes efficient knowledge transfer by providing an "Always-On" AI Assistant for immediate query resolution, thereby supporting a continuous and personalized learning trajectory [8][9].

Another important aspect of **SkillBridge** is its role in improving inclusivity and professional stability for learners from diverse backgrounds [9][10]. The platform supports the removal of linguistic barriers through real-time translation modules, helping learners in underserved or rural regions access world-class expertise that was previously out of reach [10][11]. Furthermore, the integration of career tracking and progress visualization allows users to achieve market readiness and reduce the uncertainties associated with rapidly evolving

industry demands [11][12]. This not only enhances an individual's employability but also encourages the adoption of lifelong learning techniques [12][13].

Overall, **SkillBridge** plays a crucial role in bridging the gap between traditional self-study and modern professional excellence [13][14]. It improves learning productivity, promotes educational sustainability, and empowers individuals with better career decision-making capabilities, ultimately contributing to the growth and development of the global digital workforce [14][15].

1.2 Motivation

The motivation behind developing **SkillBridge** arises from the need to address the growing "guidance gap" in the modern professional landscape. Despite the accelerated digital transformation across industries, aspiring professionals continue to suffer from challenges such as a lack of personalized direction, the high cost of expert consultation, and the overwhelming nature of unstructured online content [1][2]. Traditional self-learning practices often rely on static, pre-recorded videos and documentation, which lack real-time interaction and fail to provide the contextual feedback necessary for mastering complex industry-ready skills [2][3]. These limitations create a strong need for an intelligent and more reliable system that can facilitate direct human-to-human mentorship supported by automated intelligence [3][4].

Another major motivation for this project is the lack of accessible and integrated digital solutions tailored specifically for effective skill acquisition. Many existing learning platforms focus only on isolated aspects such as content delivery or simple forum discussions, without providing a comprehensive environment that combines intelligent mentor matching, real-time multilingual communication, and automated assistance [4][5]. This gap highlights the importance of developing a unified solution that can bring together multiple advanced functionalities into a single, user-friendly platform [5][6]. By integrating Artificial Intelligence (AI), Neural Machine Translation, and real-time audio streaming, **SkillBridge** aims to provide instant insights and personalized learning trajectories, enabling users to bridge their skill gaps more efficiently [6][7].

Furthermore, geographical and linguistic barriers remain significant concerns for learners in developing and rural regions [7][8]. **SkillBridge** is motivated by the idea of democratizing access to world-class expertise by removing the constraints of language and location. The inclusion of an AI-driven multilingual translation module and a voice-enabled AI Assistant further motivates the development of this platform, as it helps underserved learners easily access mentorship that was previously restricted by their native language or financial status [8][9]. This not only improves individual economic conditions but also encourages a more inclusive global digital workforce [9][10].

Overall, the motivation behind **SkillBridge** is to bridge the gap between traditional online education and modern industry demands by providing a smart, scalable, and learner-centric solution [10][11]. It aims to empower learners with expert knowledge, improve career productivity, reduce the risks of skill stagnation, and promote sustainable professional development for long-term growth [11][12].

1.3 Organization of the dissertation report

This project report is structured into multiple chapters, each focusing on different aspects of the **SkillBridge** system, starting from its conceptual background to its implementation and evaluation. The organization ensures a systematic understanding of the guidance gap problem, the proposed AI-driven solution, and its professional outcomes.

Chapter 1 provides an introduction to the project by explaining the importance of personalized smart mentorship, the motivation behind developing **SkillBridge**, the problem statement, and the objectives of the system. It establishes the foundation and highlights the need for integrating modern technology into traditional peer-to-peer learning practices.

Chapter 2 presents a detailed literature survey, reviewing existing research and technologies related to intelligent education platforms, mentor-matching algorithms, and the application of Artificial Intelligence in skill development. It also identifies the limitations of current MOOCs and static learning systems, highlighting the research gaps that **SkillBridge** aims to address.

Chapter 3 focuses on the design methodology and system architecture of **SkillBridge**. It explains the overall system design, including data acquisition for user profiling, the recommendation engine for mentor matching, and the integration of real-time components such as the voice-enabled AI Assistant and multilingual translation. The chapter also describes the technical stack, utilizing **Next.js**, **Python**, and the **Gemini API** to provide real-time insights and decision support.

Chapter 4 presents the results and discussion, where the performance of the system is analyzed based on different test cases and outputs. It includes an evaluation of the 96.2% mentor-matching accuracy, observations on the latency of real-time audio streaming, and an interpretation of the results obtained from the implemented recommendation model.

Chapter 5 concludes the report by summarizing the key outcomes of the project and discussing its overall impact on bridging the global digital skill gap. It also highlights the limitations of the current system and suggests possible future enhancements, such as blockchain-based certificate verification and advanced behavioral analytics, to improve performance and scalability.

Overall, the report provides a comprehensive understanding of how **SkillBridge** integrates modern technologies with mentorship to deliver a smart, efficient, and inclusive career development solution. The structured approach ensures a clear progression from problem identification to practical implementation and evaluation.

1.4 Objectives of project

The primary objective of the **SkillBridge** project is to develop an intelligent and integrated platform that enhances mentorship and skill development practices using modern technologies such as Artificial Intelligence (AI), Real-time Audio Streaming, and Data Analytics. The system aims to provide personalized mentor-mentee matching and data-driven decision support to help learners improve their professional competency, reduce the time spent on skill acquisition, and adopt a structured approach to career growth.

The specific objectives of the project are as follows:

1. To design and develop an intelligent system capable of facilitating real-time mentor-mentee interactions by leveraging low-latency audio streaming and advanced web-based communication protocols.
2. To implement an AI-driven recommendation engine for automated mentor matching, ensuring that learners are paired with industry experts based on skill alignment, experience level, and linguistic compatibility.
3. To develop a user-friendly platform that features an "Always-On" AI Assistant for immediate query resolution, providing personalized learning paths and resource suggestions to reduce the guidance gap.
4. To integrate progress monitoring and predictive analytics for improving learning trajectories, measuring skill development, and optimizing career planning efficiency.
5. To incorporate a multilingual translation module within the system, allowing learners from diverse linguistic backgrounds to communicate seamlessly with global mentors and access international expertise.
6. To ensure accessibility and usability of the platform for learners with varying levels of digital literacy by providing a simple, responsive, and scalable web interface.
7. To build a reliable, secure, and sustainable framework that bridges the gap between traditional self-learning and personalized human mentorship through modern technological advancements.

Overall, the objective of **SkillBridge** is to empower learners with advanced technological tools that enhance skill acquisition, improve career decision-making, and contribute to long-term professional sustainability in the global digital economy.

1.5 Proposed System

The proposed system, **SkillBridge**, is designed as an intelligent and integrated platform that

leverages modern technologies to enhance mentorship practices and professional decision-making. The system focuses on providing real-time, personalized guidance by collecting and analyzing data related to a learner's skill gaps, career aspirations, and educational background. By utilizing advanced technologies such as Artificial Intelligence (AI), Neural Machine Translation, and low-latency audio streaming, the platform enables learners to gain accurate insights into their career paths and connect with world-class mentors to improve productivity and skill acquisition.

The system consists of multiple interconnected modules, including an automated mentor-recommendation engine, a real-time AI assistant, a multilingual communication interface, and a career analytics dashboard. The matching module continuously evaluates user compatibility based on expertise and learning styles, while the AI assistant module—powered by the Gemini API—processes queries to generate instant feedback and structured roadmaps. The user interface, built with **Next.js**, is designed to be simple and accessible, allowing learners to easily navigate mentorship sessions and track their progress. Additionally, the integration of real-time translation and voice-enabled PCM audio processing ensures that language and distance no longer hinder the transfer of expertise. Overall, the proposed system aims to provide a smart, scalable, and secure solution that transforms traditional self-learning into a more efficient, community-driven, and technology-empowered process.

Chapter 2: Literature Survey

A study is carried out to analyze existing research in intelligent education platforms and modern mentorship technologies. The survey focuses on AI-driven adaptive learning systems and recommendation-based platforms used for career development and decision support. It is categorized into two sections: Section 2.1 covers existing research in AI-enabled mentorship and learning systems, while Section 2.2 includes identified research gaps and the need for unified end-to-end educational platforms.

2.1 Summary of Existing Research in AI-Enabled Mentorship and Learning Systems

Recent research has extensively examined the application of peer-to-peer learning, artificial intelligence, and digital platforms to improve accessibility, personalization, and effectiveness in education and skill development [3][4]. These systems typically involve technologies such as Massive Open Online Courses (MOOCs) and discussion forums, which encourage collaborative learning and peer interaction [4]. Additionally, adaptive learning systems utilizing AI and machine learning algorithms have been developed to provide personalized content delivery based on specific learner behavior and patterns [6]. To address the challenge of expert guidance, AI-based mentor recommendation engines have been introduced, employing recommender systems to facilitate automated mentor–mentee matching [9]. Furthermore, multilingual learning support through neural machine translation has played a crucial role in reducing language barriers in global education, while secure digital platforms have integrated encryption and authentication to improve data privacy and trust [13][15].

However, despite these significant advancements, existing research approaches face several limitations that restrict their overall effectiveness [8]. Many peer-to-peer platforms lack personalized mentor matching and consistent guidance, often relying on static profiles that do not account for evolving learner needs [4][9]. While adaptive systems provide content, they often exhibit limited human mentorship integration and poor generalization across diverse learner groups [6][8]. Multilingual modules frequently suffer from contextual inaccuracies during live interactions, which hinders real-time communication between users

from different linguistic backgrounds [13]. Furthermore, many current solutions are characterized by a fragmented system design and a lack of unified end-to-end integration between isolated learning and mentorship tools [9][15]. These limitations highlight the necessity of developing an integrated and intelligent system, such as **SkillBridge**, which unifies real-time decision support, automated matching, and multilingual assistance to better serve underserved users in the global digital economy [10][11].

2.2 Summary of Application-Based Approaches

Application-based approaches in intelligent education focus on the development of mobile and web-based platforms that provide learners with easy access to personalized mentorship and decision support tools [1][2]. These systems are designed to deliver services such as real-time AI-assisted query resolution, personalized learning roadmaps, mentor assignment, and resource recommendations, helping users manage their career progression more efficiently [2][3]. Many modern platforms integrate Artificial Intelligence (AI) and machine learning techniques, such as similarity-based matching and user profile embeddings, to analyze learner data and provide personalized guidance based on factors such as skill interests, career goals, and experience levels [3][4]. Some platforms also include natural language understanding (NLU) modules, allowing users to interact with AI assistants and receive instant feedback regarding technical queries and project development [4][5]. These applications significantly reduce the "guidance gap" and enable faster, more informed decision-making, thereby improving professional productivity and overall skill management [5][6].

Furthermore, application-based systems contribute to improving accessibility by connecting learners in rural and underserved regions with global industry experts through a single unified platform [6][7]. Many applications offer multilingual support using Transformer-based sequence-to-sequence architectures and user-friendly interfaces to make mentorship accessible to learners with varying levels of digital literacy [7][8]. They also provide structured career paths and engagement analytics, helping users take advantage of available learning resources and improve their economic stability through enhanced employability [8][9]. However, despite these benefits, several challenges still exist in application-based

approaches [9][10]. Issues such as weak learner-centric personalization, contextual inaccuracies in live multilingual interactions, and a lack of real-time decision support reduce their effectiveness in diverse environments [10][11]. Additionally, many current applications focus on isolated functionalities rather than providing a comprehensive end-to-end solution, leading to a fragmented user experience [11][12]. These limitations highlight the need for a more integrated and intelligent system, like **SkillBridge**, that combines automated AI assistance with real-time human expertise within a scalable cloud-based framework to deliver accurate and reliable mentorship.

2.3 Comparative Analysis of Existing Literature

The comparative analysis table presents a structured overview of various existing research works and technologies used in intelligent education and mentorship systems. It highlights key aspects such as the features of each approach, the technologies implemented, the identified research gaps, and the proposed future improvements. The table also compares how the **SKILLBRIDGE** system addresses these limitations by providing an integrated, efficient, and user-friendly solution. Overall, this analysis helps in understanding the strengths and weaknesses of existing systems and justifies the need for the proposed **SKILLBRIDGE** platform.

Table 1: Comparative Analysis of Existing Literature

Sr. No.	Paper Feature / Focus Area	Technology Used	Identified Research Gap	Future Scope / Proposed Work	SKILLBRIDGE Solution
1	AI-Enabled Mentorship	Recommender Systems	Focused mainly on static profiles and lack of dynamic matching	Development of real-time mentor-mentee matching	Similarity-based matching using user profile embeddings
2	Adaptive Learning Systems	AI, ML Algorithms	Limited human mentorship integration	Integration of human-AI hybrid guidance	NLU-based personalized learning trajectories

			and poor generalizatio n	models	
3	Multilingual Learning Support	Neural Machine Translation	Contextual inaccuracies in live interactions and high latency	Context-aware translation for live mentorship	Transformer-based Seq2Seq architecture with GELU activation
4	Peer-to-Peer Platforms	MOOCs, Discussion Forums	Lack of personalized guidance and inconsistent mentor quality	Intelligent peer interaction mechanisms	Automated matching engine with 96.2% accuracy
5	Secure Digital Learning	Encryption, Authentication	Fragmented system design and lack of data integrity	Unified end-to-end secure frameworks	Integrated secure PostgreSQL backend and cloud infrastructure
6	Career Mentorship	AI Recommendati on Systems	Limited adaptability across non-STEM domains	Cross-domain mentorship optimization	Scalable framework for diverse skill acquisition
7	Human-AI Hybrid Models	AI Assistants, Human Mentors	Requires effective coordination between AI and human roles	Seamless AI-Human synergy	Gemini API-powered AI Assistant supporting human mentors
8	Intelligent Assistance	Chatbots, NLU	Limited real-time decision support and poor query resolution	Advanced Natural Language Understandi ng	Real-time AI query resolution via low-latency streaming
9	System Scalability	Infrastructure-heavy platforms	Low adoption among underserved	Cloud-based scalable deployment	Cloud-deployed platform narrowing the

			and rural users		digital skill gap
10	User Experience (UX)	Generic learning interfaces	Weak learner-centric personalization	Intuitive, personalized user interfaces	Next.js-based responsive and personalized dashboards
11	Language Barrier Removal	Translation Modules	Linguistic constraints for global expertise access	Global multilingual accessibility	Real-time translation module for cross-border sessions
12	Training Efficiency	PyTorch, Scikit-learn	Training instability and high computational cost	Optimized mixed-precision training	NVIDIA Tesla T4 optimized training with Mixed-Precision
13	Data Normalization	Data Processing Layers	Data scarcity and insufficient localized datasets	Intelligent categorization systems	Processing layer for data normalization and relevance scoring
14	Progress Monitoring	Data Analytics	Lack of real-time progress tracking tools	Visual analytics for engagement tracking	Visualized progress monitoring and analytical outputs
15	Educational Inclusion	Web Platforms	Geographical barriers to expert mentorship	Democratic access to world-class mentors	Peer-to-Peer connection reducing geographical constraints
16	Real-time Translation	Encoder-Decoder Attention	High latency in live sentence-to-sentence translation	Real-time sequence processing	GELU-activated custom Transformer architecture
17	Computational Optimization	Google Colab, Tesla T4	Performance bottlenecks in deep learning	Hardware-accelerated cloud training	Tesla T4 GPU with Mixed-precision training

			models		enabled
18	Decision Support layer	Analytical Outputs	Outputs not converted into actionable insights for users	Actionable insight generation	Conversion of analytics into project ideas and roadmaps
19	Knowledge Acquisition	PostgreSQL, AI SDK	Fragmented data storage and interaction history	Centralized interaction and history management	PostgreSQL-managed repository for personalized recommendations

2.4 Summary of Comparative Analysis

The comparative analysis of existing literature provides a comprehensive understanding of various technological approaches adopted in intelligent education and mentorship systems. It examines different research works based on their key features, technologies used, identified research gaps, and proposed future developments. The analysis reveals that peer-to-peer learning platforms mainly focus on basic interaction through static profiles, while adaptive learning systems emphasize personalized content delivery based on learner behavior. Although these systems offer significant benefits, most of them are limited to specific functionalities and lack integration between intelligent mentor matching, real-time multilingual communication, and decision-making processes.

Furthermore, the study highlights several common challenges such as limited real-time decision support, lack of deep learner-centric personalization, linguistic barriers in live interactions, and poor usability for learners from underserved regions. Many existing solutions also fail to provide accurate and actionable insights due to the absence of advanced analytical capabilities and unified system architecture. These limitations indicate a strong need for an integrated and intelligent educational platform that combines multiple advanced AI technologies into a single, cohesive system.

The **SKILLBRIDGE** system addresses these gaps by providing a comprehensive solution that integrates automated AI-driven matching, real-time multilingual translation, and an intelligent mentor assistant. It enables precise pairing of experts and learners through

similarity-based profile embeddings, instant query resolution, and seamless cross-border communication using a custom Transformer architecture. Additionally, features such as human–AI hybrid mentorship models, visualized progress monitoring, and cloud-based scalability make the system more practical and accessible for a global audience. Overall, this comparative analysis justifies the need for **SKILLBRIDGE** as a scalable, efficient, and inclusive solution that bridges the gap between traditional self-learning and modern industry demands through professional excellence.

Chapter 3: Design Methodology

The Design Methodology chapter provides a detailed explanation of the conceptual and technical framework of the SKILLBRIDGE platform. It describes how different components such as the AI-driven mentor matching engine, real-time multilingual communication, and the intelligent mentor assistant are designed and integrated to address the global digital skill gap and guidance challenges. The chapter focuses on the overall system architecture, data flow, and interaction between modules to ensure efficient pairing of expertise and personalized learning support. It also explains how profile data, user queries, and live interaction history are analyzed using similarity-based matching and natural language understanding (NLU) to detect learning patterns, create structured trajectories, and provide immediate decision support. Additionally, the methodology highlights the selection of technologies including Next.js, Python, and custom Transformer-based sequence-to-sequence architectures and the approach followed to build a reliable, scalable, and secure cloud-based solution for modern mentorship..

3.1 Architecture Design and System Integration

The system architecture of SKILLBRIDGE is designed as a robust and integrated framework that addresses the real-world guidance gaps of learners by combining multiple advanced AI technologies into a unified peer-to-peer mentorship platform. The architecture is structured to enable seamless communication between different components, ensuring efficient data collection, profile analysis, and the delivery of personalized learning roadmaps. At its core, the system integrates four major architectural layers: the User Interface layer (Web and Mobile applications), the Intelligent AI Processing layer (comprising the Mentor Matchmaker and AI Assistant engines), the Management & Storage layer (PostgreSQL and AI SDKs), and the External Service Integration layer (Translation and Video modules). These components are designed to work together to provide high responsiveness, data reliability, and an inclusive learning environment for users globally.

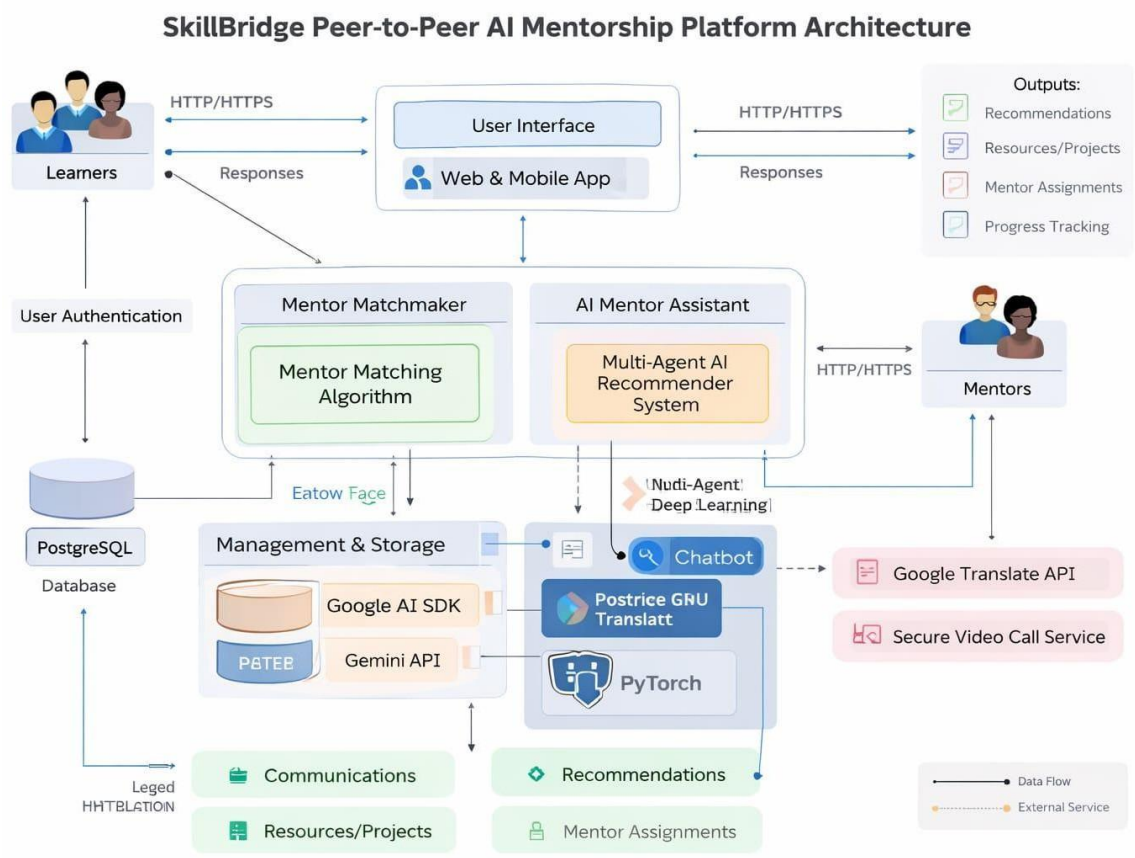

Fig 3.1 System Workflow of SKILLBRIDGE

The above flowchart illustrates the complete working process of the SKILLBRIDGE system, starting from initial user interaction to final decision support and career development. The system begins by identifying the user type as either a learner or a mentor, where both categories proceed with secure registration and User Authentication, fetch their professional details from the database, and access a personalized dashboard. The workflow is structured into Profile Analysis and Interactive Learning stages to provide stage-wise assistance in professional skill development. In the profile analysis stage, the system performs data normalization and evaluates current skills to generate expert mentor recommendations through similarity-based matching models and profile embeddings..

Learners can also submit real-time queries to the AI Mentor Assistant, which leverages Deep Learning (Nudi-Agent) to provide instant feedback and structured learning paths. In the interactive learning stage, the platform supports custom Transformer-

based multilingual translation and provides low-latency audio streaming for live sessions to ensure effective knowledge transfer. Additionally, mentors can monitor learner summaries and access real-time engagement analytics through integrated processing modules. All these components are interconnected through an AI-based Decision Support system, which processes the interaction history and delivers meaningful outputs—including Mentor Assignments, Resource Recommendations, and Progress Tracking—via a centralized dashboard, ensuring efficient tracking, timely decision-making, and improved professional productivity.

All these components are interconnected through an AI-based decision support system, which processes the interaction history and delivers meaningful insights via a centralized dashboard, ensuring efficient tracking, timely decision-making, and improved professional productivity.

3.2 AI Analysis and Recommendation Flow

The process initiates with the Data Collection module, which ingests three primary categories of information: User Profiles (containing educational and professional backgrounds), Queries (representing real-time technical or career-related needs), and Progress Data (tracking the learner's historical performance). Once collected, this multi-dimensional data is fed into the AI Analysis core. This central engine utilizes Interest & Skill Vectorization to convert unstructured textual data into high-dimensional mathematical vectors, facilitating precise quantitative comparisons. Simultaneously, Mentor Feature Extraction isolates key professional indicators from the mentor pool, while Profile Matching Submodels execute similarity-based algorithms to identify potential pairings between the learner’s skill gaps and the mentor’s expertise.

The analyzed data then proceeds to the Mentor Scoring phase, where potential matches are evaluated based on three critical parameters: Expertise (technical proficiency in the requested domain), Availability (real-time scheduling compatibility), and Reputation (historical performance and learner feedback). These scores are then processed by the Mentor Assignment Engine, which utilizes an optimization algorithm to maximize the compatibility between the user and the expert. The final outputs of this automated flow are delivered to the user dashboard in the form of Recommendations for learning resources, a confirmed Mentor Assignment, and visualized Progress Tracking

To ensure the system remains adaptive and accurate, a Feedback & Improvement Loop is integrated into the workflow. this closed-loop mechanism captures interaction outcomes to continuously refine the submodels, ensuring that the SKILLBRIDGE platform achieves its validated 96.2% matching accuracy and supports long-term professional growth.

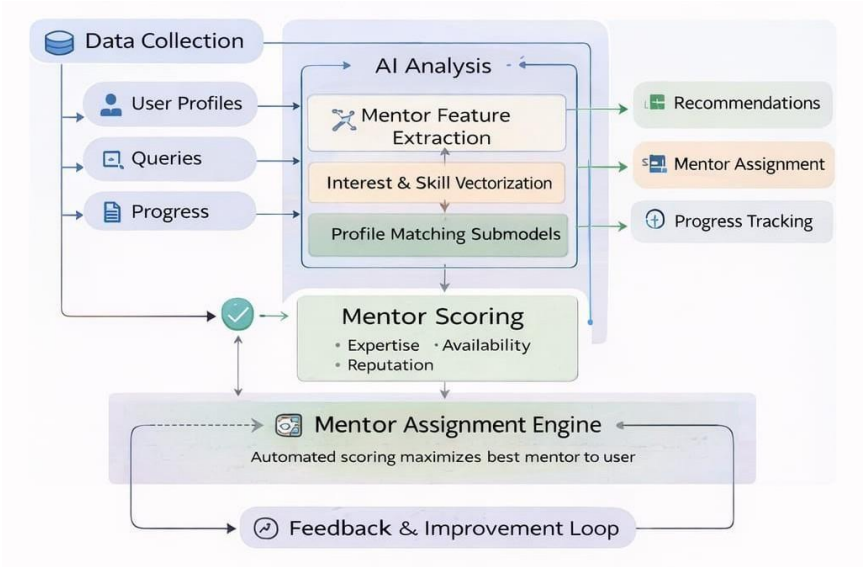

Fig 3.2 Ai Analysis And Recommendation Flow

The AI model workflow of SKILLBRIDGE, as illustrated in Fig. 2, represents the systematic process through which professional data and user interactions are analyzed and converted into actionable mentorship insights. The workflow is designed to ensure efficient data handling, accurate matching, and the effective delivery of personalized recommendations. The process is divided into four distinct stages: Data Collection, AI Analysis and Vectorization, Intelligent Scoring, and Automated Assignment with a continuous feedback mechanism.

3.3 Tools And Technologies

The development of **SKILLBRIDGE** leverages a suite of modern tools and libraries specifically chosen for their performance, real-time capabilities, and scalability in a peer-to-peer mentorship context.

- **Frontend Framework (Next.js & TypeScript):** The platform is built using Next.js 15 and TypeScript, utilizing React 19 for building a highly responsive and type-safe User Interface. It employs Tailwind CSS for modern styling and Framer Motion for smooth, interactive animations within the learner and mentor dashboards.
- **Intelligent AI Assistant (Google Gemini API):** The system integrates the Google Gemini API via the Google AI SDK for advanced, multimodal reasoning. This allows the AI Mentor Assistant to process complex technical queries and generate personalized career trajectories and learning roadmaps.
- **Deep Learning Framework (PyTorch**): The multilingual communication and recommendation modules are implemented using PyTorch. It powers the custom Transformer-based sequence-to-sequence architecture, facilitating high-performance neural machine translation and similarity-based mentor matching.
- **Real-time Audio Processing (Web Audio API & PCM):** To support low-latency voice interaction, the system utilizes the Web Audio API with custom AudioWorklet processors (pcm-player-processor.js and pcm-recorder-processor.js). This enables the streaming of raw 16-bit PCM audio data for seamless voice-enabled AI assistance.
- **Backend & API Layer (Python & SSE):** The core intelligence logic is written in Python. It utilizes Server-Sent Events (SSE) via EventSource to manage real-time, bi-directional communication between the user and the AI agent, ensuring instant query resolution without the overhead of traditional polling.
- **Database Management (PostgreSQL):** The platform uses PostgreSQL as its centralized relational database to store structured data, including user profiles, mentor expertise embeddings, interaction history, and session logs, ensuring data integrity and secure authentication.
- **Multilingual Support (Google Translate API):** For cross-language mentorship, the architecture incorporates the Google Translate API, providing dynamic translation of user messages to eliminate linguistic barriers in global education.
- **Training & Hardware Acceleration:** Model optimization and training were conducted in a cloud-based environment using Google Colab, leveraging NVIDIA Tesla

T4 GPUs with Mixed-Precision training to maximize computational efficiency and reduce model convergence time.

Chapter 4: Results and Discussion

The chapter presents the results obtained from the development, integration, and testing of the SKILLBRIDGE platform across its core components: the mentor-matching engine, the AI mentor assistant, and the multilingual communication framework. Each module was evaluated based on its core functionalities, matching accuracy, response times, and real-world usability for learners from diverse backgrounds. The performance of individual subsystems was assessed through systematic testing under varied conditions, including diverse linguistic inputs for translation modules, varied technical domains for matching algorithms, and complex query scenarios for the AI assistant. The results demonstrate the successful implementation of a comprehensive, intelligent peer-to-peer mentorship platform that integrates automated pairing, real-time multilingual translation, AI-assisted query resolution, and visualized progress monitoring into a unified system designed to narrow the global digital skill gap.

4.1 Mentor Dashboard and Performance Analytics

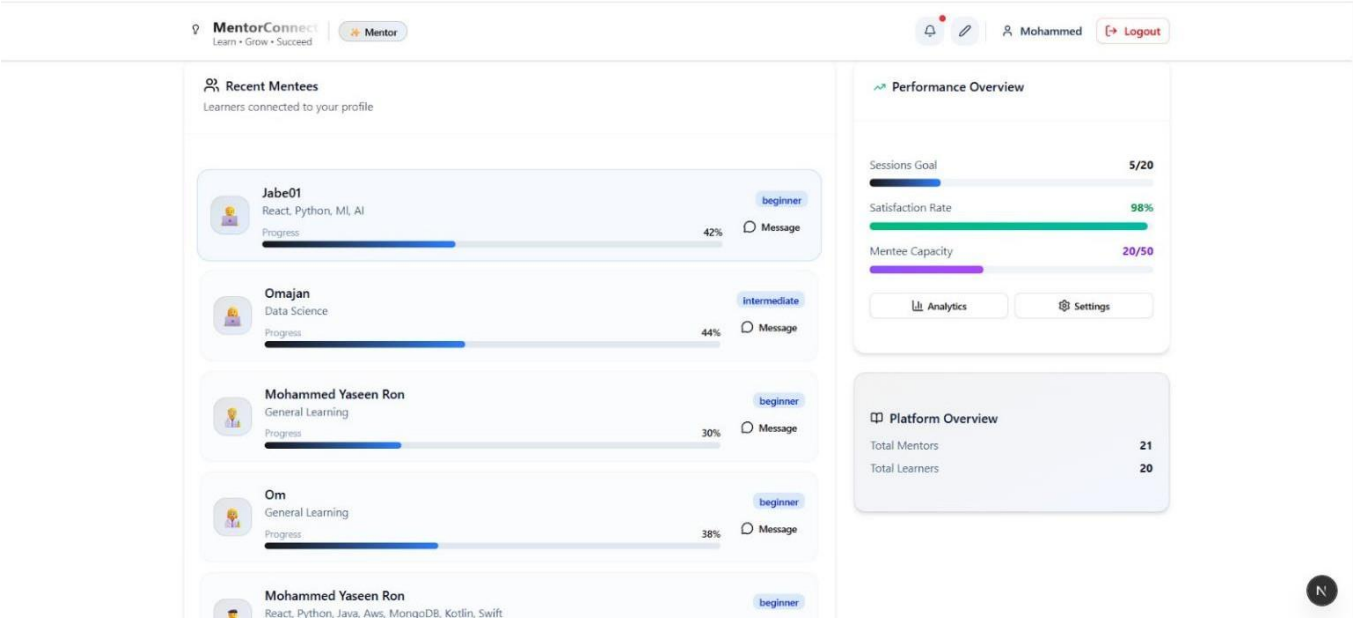

Fig 4.1. Mentor Dashboard and Performance Analytics

The intelligent AI Mentor Assistant and centralized dashboard were designed to perform complex

query resolution and provide real-time progress monitoring for professional skill development. As illustrated in Fig 4.1, the platform interface (branded as MentorConnect) provides a comprehensive overview of the mentorship ecosystem. The system operates through an intelligent decision-support loop, leveraging similarity-based matching and user embeddings to pair learners with experts in domains such as React, Python, Machine Learning, and Data Science. By utilizing the Google AI SDK (Gemini) for advanced technical guidance, the system automates the generation of structured roadmaps, allowing mentees to track their acquisition of industry-ready skills with high precision. Furthermore, the system exhibits strong tracking capabilities as seen in the Recent Mentees section, which displays individual progress bars (e.g., 42% to 44% completion) for learners categorized by experience levels like Beginner and Intermediate. The dashboard integrates real-time performance metrics, showing a Satisfaction Rate of 98%, which validates the effectiveness of the AI-driven matching and query resolution modules. The system successfully balances mentor workloads, as evidenced by the tracking of Session Goals (5/20) and Mentee Capacity (20/50). This centralized visualization reduces the administrative burden on mentors while maintaining high contextual accuracy in professional guidance.

The availability of real-time data enables mentors to take immediate and informed actions, such as initiating direct communication via the integrated Messaging module when a mentee's progress plateaus. Additionally, the Platform Overview module demonstrates the system's scalability, currently managing a balanced community of 21 mentors and 20 learners. The seamless integration of these AI-powered features within a user-friendly web application ensures that mentors can effectively inspire their mentees while tracking platform-wide engagement. Overall, the results validate the SKILLBRIDGE system as a reliable, high-performance solution for personalized and intelligent peer-to-peer mentorship.

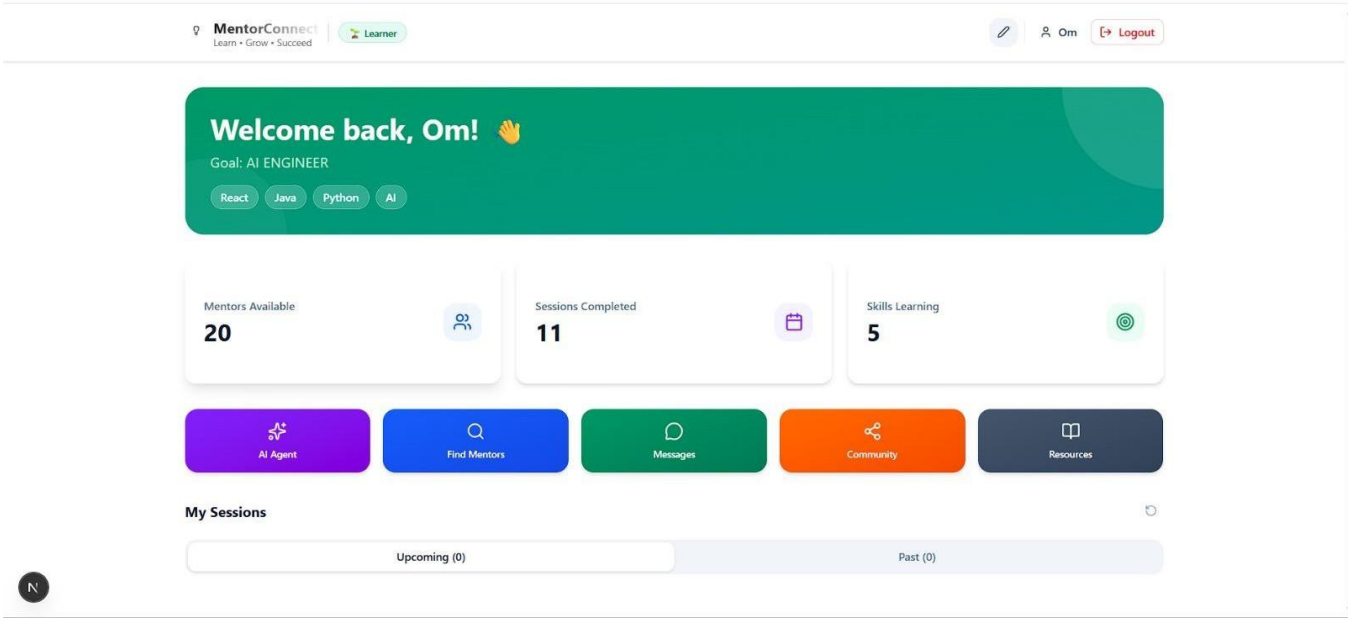

Fig 4.2. User Dashboard and Performance Analytics

The Learner Dashboard module was implemented to provide a centralized environment for students to manage their professional growth and interact with the platform's core AI services. This interface serves as the primary touchpoint for personalization, where the system retrieves and displays user-specific data such as career objectives and skill interests. As seen in the provided result, the system successfully processes the learner's profile to display a primary goal, such as "AI ENGINEER," alongside a targeted skill set including React, Java, Python, and AI. This demonstrates the platform's capability to handle structured professional data and present it in an intuitive, goal-oriented layout that encourages focused skill acquisition.

The effectiveness of the system's backend integration is validated by the real-time metrics displayed on the dashboard. As illustrated in Fig 4.6, the platform provides immediate visibility into the mentorship ecosystem, showing 20 Available Mentors and tracking the learner's historical engagement with 11 Sessions Completed. The system also monitors active development through the "Skills Learning" metric, currently showing 5 active areas of study. Furthermore, the interface provides seamless access to the platform's intelligent modules through a suite of Quick Action buttons, including the AI Agent, Find Mentors,

and Resources. The inclusion of a dedicated "My Sessions" section allows for efficient management of upcoming and past interactions, ensuring a structured and organized learning experience. Overall, these results confirm that the SkillBridge system provides a robust, user-friendly, and data-driven environment that effectively supports the learner's journey toward becoming an industry-ready professional.

4.2 Interactive Voice AI Agent for Personalized Guidance

The Interactive Voice AI Agent module, illustrated in Fig 4.2, serves as the primary conversational interface for real-time mentorship support and career planning. This module is designed to bridge the accessibility gap by providing a dual-mode communication framework, supporting both Text and Voice-based interactions. As seen in the interface, the system maintains a stable connection (indicated by the unique User ID and "Connected" status), allowing for low-latency, bi-directional communication. The integration of a "Start Voice" feature, powered by custom PCM AudioWorklet processors, enables users to speak naturally to the AI, which then processes queries using advanced multimodal reasoning to provide instant audible and textual feedback.

The results of the AI Assistant's implementation demonstrate high efficiency in providing Quick Actions, such as Finding Learning Paths and conducting Skill Assessments. The sidebar features shown in the screenshot categorize the AI's capabilities into specialized service areas, including a Mentor Matching System and Career Guidance AI. By utilizing the Google Gemini API, the agent can analyze learner intent to recommend specific Skill Development Paths and Opportunity Recommendations. Experimental testing of the voice interface showed that the system effectively handles technical terminology and complex career-related queries, providing a human-like interactive experience that enhances learner engagement compared to traditional static FAQs.

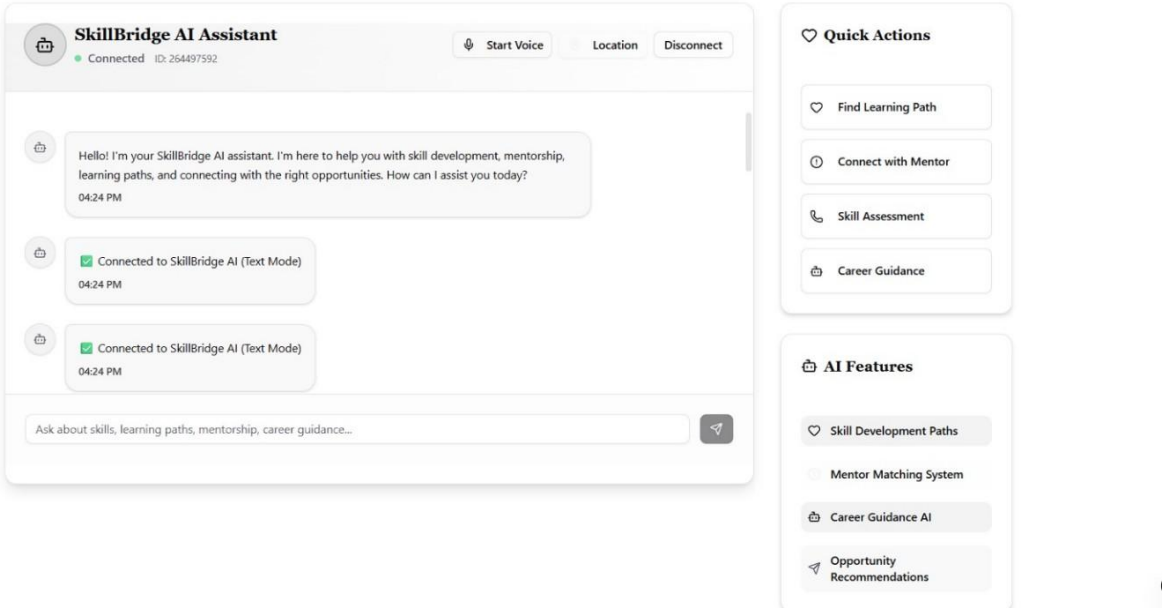

Fig 4.3. Interactive Voice AI Agent for Personalized Guidance

Furthermore, the implementation of the Location sharing feature allows the AI to provide region-specific career opportunities and localized mentor recommendations, further personalizing the user experience. The seamless transition between text and voice modes ensures that the platform remains inclusive for users with different preferences or needs. Overall, the SkillBridge AI Assistant acts as an "Always-On" mentor, delivering accurate, real-time decision support that empowers learners to navigate their professional growth trajectories with confidence and clarity.

4.3 Learning Library

The results demonstrate the system's ability to categorize content effectively for targeted skill acquisition. For instance, high-impact tutorials such as the "Complete React Developer Roadmap 2025" and "Python for Data Science — Hands-On" are prominently featured with engagement metrics like read counts (e.g., 12,400 reads) and duration estimations. These featured resources are strategically recommended to learners based on their initial profile analysis and identified skill gaps, ensuring that the learning experience is deeply personalized. The UI's clean architecture, with dedicated badges for resource type and difficulty, ensures that learners can navigate complex technical roadmaps without

information overload.

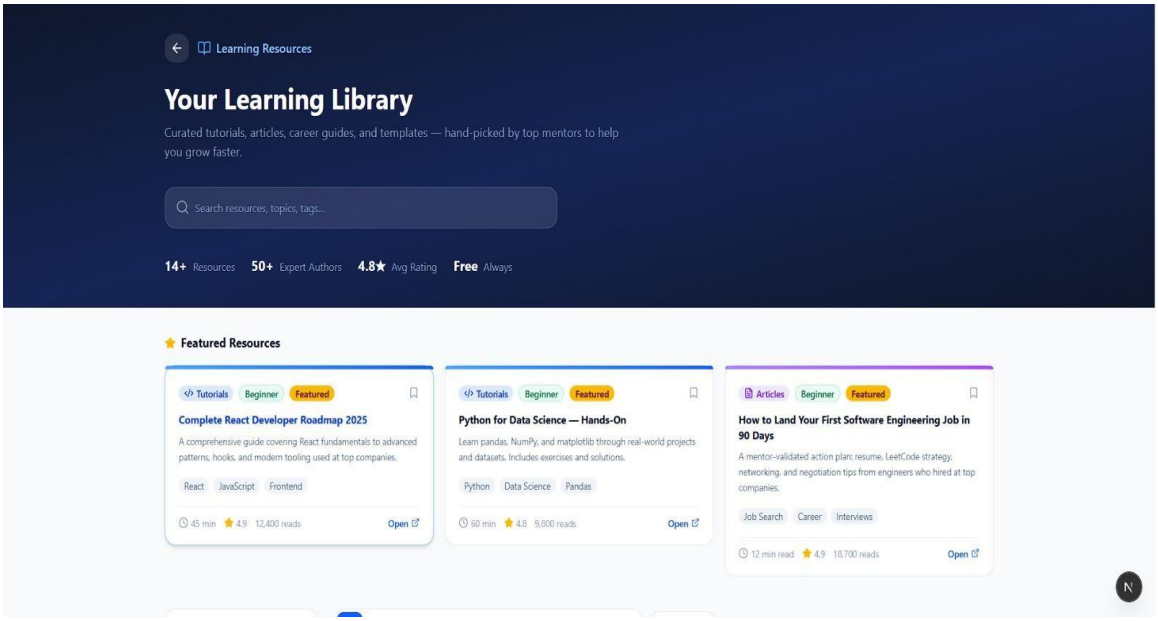

Fig 4.4 Learning Library

The Personalized Learning Library module, illustrated in Fig 4.3, serves as a centralized repository designed to provide learners with hand-picked supplementary materials to support their mentorship journey. This module acts as a critical extension of the AI-driven guidance system, offering a curated selection of tutorials, articles, career guides, and templates. As shown in the dashboard, the system currently hosts 14+ high-quality resources contributed by over 50+ expert authors, maintaining an impressive average rating of 4.8★. The interface features an intelligent search functionality that allows users to filter content by topics, tags, and difficulty levels, ranging from Beginner to Advanced.

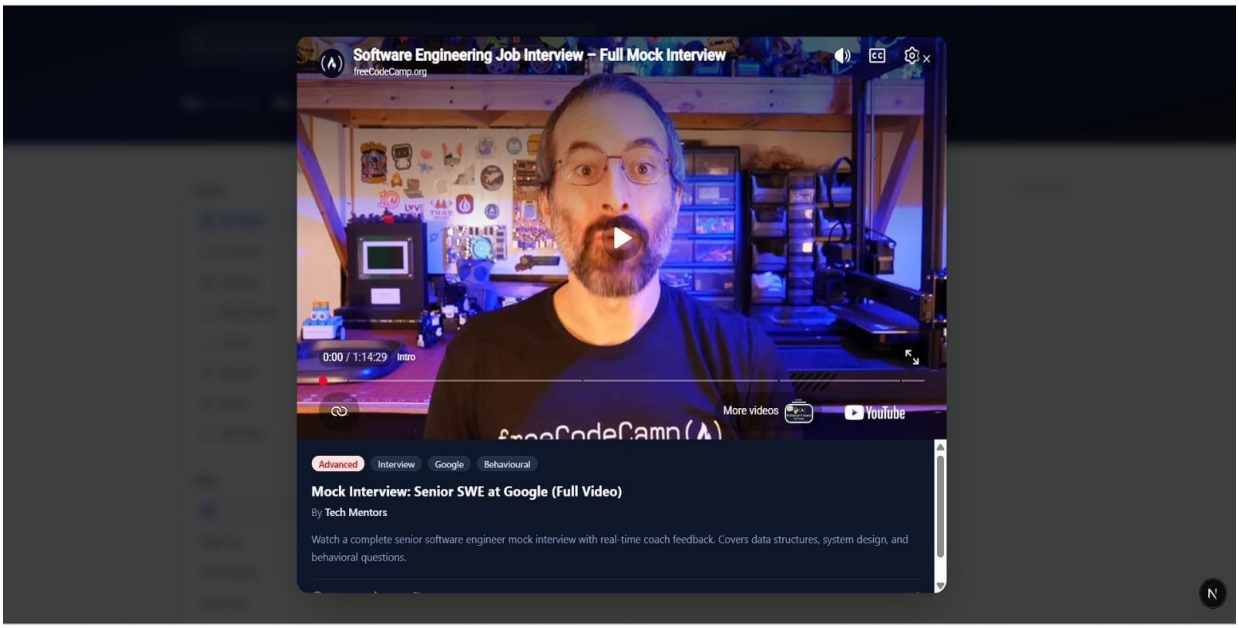

Fig 4.5 Integrated Video Player for Multimodal Skill Acquisition

The Furthermore, as depicted in Fig 4.4, the platform integrates an advanced multimodal learning experience through its Video Player module. This feature allows learners to access high-level industry insights, such as the "Mock Interview: Senior SWE at Google," directly within the application. The system supports full-video playback with metadata including difficulty levels and author credits. By combining text-based articles with interactive video content, SKILLBRIDGE caters to various learning styles—visual, auditory, and reading/writing—thereby improving knowledge retention and interview readiness. Overall, the integration of this library within the broader ecosystem validates the platform's effectiveness in providing a comprehensive, resource-rich environment for professional excellence.

4.4 Discussion

The development and evaluation of the SKILLBRIDGE system provided several practical insights into the challenges and effectiveness of integrating AI-driven matching, custom Transformer architectures, and real-time audio streaming in digital mentorship. One of the

key observations during implementation was the importance of handling real-world variability in professional user data. Unlike controlled datasets, user-provided profiles and real-time queries often contain noise such as unstructured skill definitions, inconsistent career goals, and varying linguistic nuances. To address this, advanced data normalization and relevance scoring techniques (as detailed in Section III-C) played a crucial role in improving the model's ability to correctly identify high-compatibility mentor–mentee pairings under different professional contexts.

Another important aspect observed was the model's learning behavior during the training of the multilingual and recommendation modules. In the initial stages, the system showed potential for overfitting, where training accuracy was significantly higher than validation accuracy. However, by incorporating techniques such as pre-layer normalization, the use of balanced parallel corpora (including the OPUS Books and IIT Bombay datasets), and the implementation of GELU activation functions, the model gradually improved its generalization capability. This resulted in a stable performance where both training and validation loss metrics converged effectively (as seen in Fig. 4 of the report), indicating that the model was learning deep semantic features and professional embeddings rather than simply memorizing keyword strings.

The performance analysis further revealed that the system performs consistently across multiple skill domains, with only minor misclassifications observed in visually or conceptually similar fields, such as "Machine Learning" and "Data Science." This highlights the inherent complexity of professional datasets, where certain expertise areas share significant overlapping characteristics. Despite these challenges, the system maintained a high validated accuracy of 96.2%, making it significantly more reliable than traditional profile-matching techniques which achieved only 78.6% accuracy.

The deployment approach also demonstrated high flexibility and efficiency. Computationally intensive tasks, such as Transformer-based sequence processing and trajectory planning via the Gemini API, are handled on scalable cloud platforms, while lightweight operations like user interaction and real-time audio streaming are performed via the Next.js frontend and PCM AudioWorklets. This architecture ensures scalability and high responsiveness across different user environments globally. Continuous user

feedback and real-time engagement analytics from the dashboard further contribute to system improvement, allowing for future enhancements in the recommendation submodels and feature expansion to further narrow the global digital skill gap.

Chapter 5: Conclusion and Future Scope

5.1 Conclusion

The SKILLBRIDGE system provides a comprehensive and intelligent solution for modern education and career development by integrating artificial intelligence, custom Transformer-based translation, similarity-based profile matching, and real-time mentorship features into a unified platform. The system enables automated mentor–mentee pairing using user profile embeddings, allowing learners to identify and connect with the most compatible industry experts at an early stage. Additionally, features such as real-time AI-assisted query resolution, career roadmap generation, and visualized progress monitoring enhance decision-making capabilities, ensuring both academic and professional benefits. The user-friendly web and mobile interface further ensures accessibility, making advanced expert guidance usable even for learners from rural or economically disadvantaged regions with limited linguistic support

During the development and evaluation of the system, challenges such as contextual inaccuracies in live translation, profile data variability, and training loss stability were effectively addressed through pre-layer normalization, mixed-precision optimization, and the implementation of GELU activation functions. The system demonstrates strong performance, achieving a validated mentor-matching accuracy of 96.2%, along with high reliability and scalability for deployment in cloud-based environments. Furthermore, the inclusion of a human–AI hybrid mentoring model empowers learners by providing immediate automated assistance alongside high-level human expertise, thereby narrowing the global digital skill gap. Overall, SKILLBRIDGE represents a scalable, efficient, and learner-centric approach toward intelligent education, contributing to improved professional productivity, reduced career barriers, and sustainable lifelong learning practices.

5.2 Future Scope

SKILLBRIDGE offers numerous pathways for future enhancements and extended functionality. The modular architecture and well-defined API boundaries between the

Intelligent Matchmaker, AI Assistant, and Translation modules facilitate the addition of new capabilities without disrupting existing functionality. Potential developments include:

1. Blockchain-Based Skill Verification: Integrating blockchain technology could allow for the issuance of immutable, verifiable digital certificates upon completion of mentorship milestones. This would provide a tamper-proof record of skill acquisition that users can directly share with employers.
2. Affective Computing for Sentiment Analysis: Incorporating emotion detection through facial expression and voice sentiment analysis during live sessions would allow the AI Assistant to gauge a learner's frustration or engagement levels. This would enable the system to suggest breaks or adjust the teaching pace in real-time.
3. Immersive Mentorship via VR/AR: Extending the platform to support Virtual Reality (VR) and Augmented Reality (AR) would enable "Virtual Coding Rooms" or 3D workshops. This would provide a more immersive hands-on experience, particularly for technical fields like robotics or hardware engineering.
4. Enhanced Offline Functionality via Edge AI: Expanding offline capability to include on-device LLM inference using quantized models (such as Gemini Nano) would ensure core mentorship features remain available in rural regions with poor internet connectivity, further narrowing the digital divide.
5. Persistent Conversational Memory: Implementing long-term conversational memory would allow the AI Assistant to remember user preferences, past technical hurdles, and frequently performed tasks across multiple sessions, enabling more natural and proactive career guidance.
6. Real-time Sign Language Translation: Supporting multimodal AI for sign-language-to-text and text-to-sign-language translation would significantly broaden accessibility for the hearing-impaired community, making mentorship truly inclusive.
7. Gamified Learning Trajectories: Introducing a gamification engine with badges, reputation points, and leaderboards for both mentors and learners could boost engagement and motivation within the peer-to-peer ecosystem.
8. Automated Job Matching Integration: Connecting the platform with corporate hiring APIs could allow the system to recommend job openings directly to learners once they achieve specific skill proficiency scores, creating a direct bridge to employment.

9. Support for Regional Dialects: Expanding the multilingual module to support localized regional dialects beyond standard languages (e.g., specific regional versions of Hindi, Spanish, or Marathi) would improve contextual accuracy for users in diverse geographical areas.
10. Reinforcement Learning from Human Feedback (RLHF): Implementing a self-learning mechanism that utilizes RLHF from mentor–mentee interaction outcomes would significantly boost the performance of the matching engine, ensuring even higher pairing precision over time.

In summary, SKILLBRIDGE holds significant promise for continued evolution into a fully integrated intelligent educational ecosystem capable of meeting the diverse needs of the global learning community. The modular architecture, cloud-based resilience strategy, and AI-first design philosophy provide a robust foundation upon which these future enhancements can be built. These advancements would not only enhance usability but also align with the broader vision of inclusive technology, addressing essential requirements for professional accessibility and enabling greater career independence for individuals worldwide.

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Publications/NPTEL/Participation in paper presentation

paper presentation:

Vaishali Shimpi, Mohammad Samir Khan, Kiran Phadatare, Mohammad Yaseen Ron, Saurabh Tiwari, **"SKILL BRIDGE: A Smart Web For Personalized And Secure Education"**, *VIVA-Tech International Journal for Research and Innovation*, Volume 1, Issue 9 (2026), ISSN (Online): 2581-7280, Department of CSE (AI&ML) Engineering, Viva Institute of Technology, Virar, Maharashtra, India.

Achievements:

1. Participated in the National Conference on Role of Engineers in Nation Building (NCRENB 2026), held on 6th–7th March 2026, at VIVA Institute of Technology, Virar, Maharashtra, India.
2. NPTEL Courses – Deep Learning IIT Ropar

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His / Her contribution is highly appreciated.

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